Droplet morphology in turbulence

TI2025, La Palma, Spain, 31 March 2025

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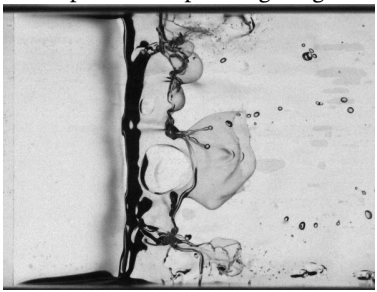
Morphology of clean and surfactant-laden droplets in turbulent flows

CO_2 absorption[a]



Wave breaking on the coast of Okinawa[b]

Droplet breakup during cough[c]



1mm water & glycerol film with 15m/s airflow.

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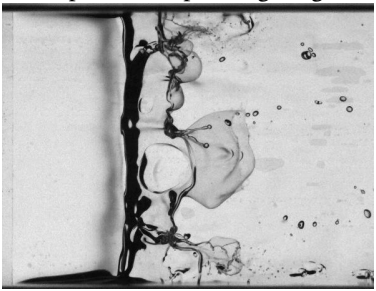
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CO_2 absorption[a]



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1mm water & glycerol film with 15m/s airflow.

What is the morphology of droplets in turbulence?

How do surfactants affect morphology?

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Numerical method

We use the in-house solver *Fujin*.



Fluid motion[d],

$$\rho\frac{\partial \mathbf{u}}{\partial t} + \rho\mathbf{u}\cdot\nabla\mathbf{u} = -\nabla p + \nabla\cdot\left[\mu\left(\nabla\mathbf{u} + \nabla\mathbf{u}^T\right)\right] + \nabla\cdot(\tau_c f_\sigma) + \mathbf{f_s},$$

$$\nabla\cdot\mathbf{u} = 0, \tag{1}$$

Droplet motion[e],

$$\frac{\partial \phi}{\partial t} + \nabla\cdot(\mathbf{u}\phi) = 0, \tag{2}$$

Surfactant motion[f],

$$\frac{\partial \psi}{\partial t} + \mathbf{u}\cdot\nabla\psi = \nabla\cdot(M_\psi\nabla\mu_\psi), \tag{3}$$

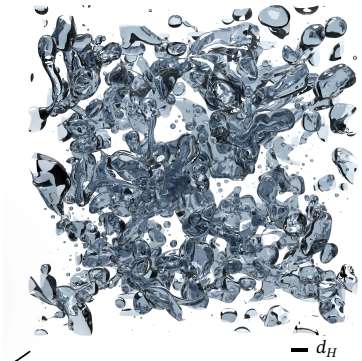
Surface tension[g],

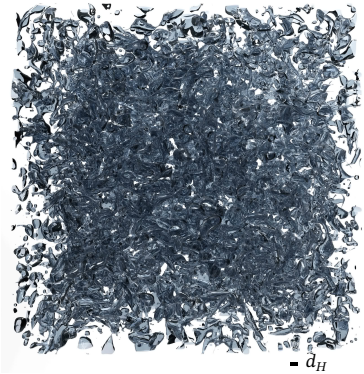
$$f_\sigma = \max[\sigma_{min}, \sigma_0(1+\beta_s\log(1-\psi))]. \tag{4}$$

[d] Eswaran, V., & Pope, S. B. (1988). *Computers & Fluids*, (3).
[e] Ii, S., Sugiyama, K., Takeuchi, S., Takagi, S., Matsumoto, Y., & Xiao, F. (2012). *Journal of Computational Physics*, (5).
[f] Engblom, S., Do-Quang, M., Amberg, G., & Tornberg, A. (2013). *Commun. Comput. Phys.*, (4).
[g] Muradoglu, M., & Tryggvason, G. (2008). *Journal of Computational Physics*, (4).

Setup

- Tri-periodic domain with $500 \times 500 \times 500$ grid points.
- Six simulations with three values of surface tension, and two surfactant concentrations.
- Hinze-scale[h] $d_H \equiv 0.725\sigma^{3/5}\rho^{-3/5}\epsilon^{-2/5}$.



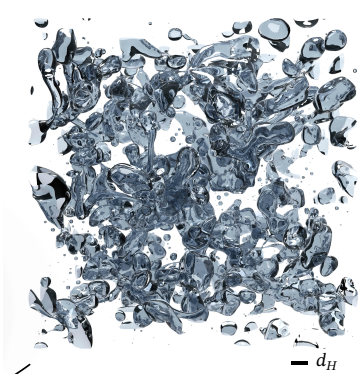


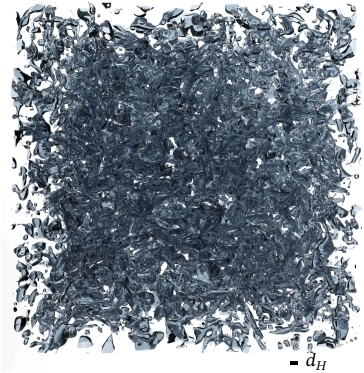
case		$\langle\phi\rangle$	$\langle\psi\rangle$	We
SP	●	0	0	-
C10	●	0.1	0	10
C20	●	0.1	0	20
C40	●	0.1	0	40
S05	◆	0.1	0.1	5
S10	◆	0.1	0.1	10
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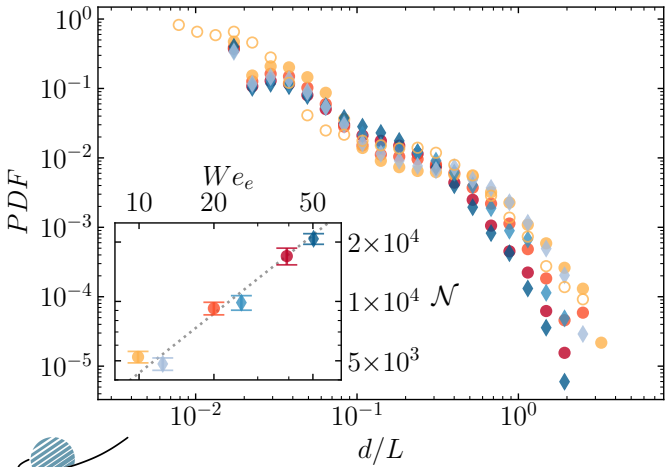




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Counting droplets



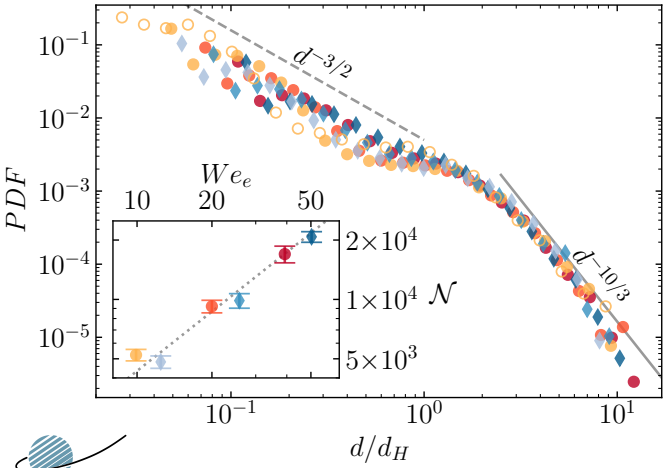
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- We use a flood-fill algorithm to count the droplets[a].
- Surface tension $\sigma \downarrow$, number of droplets $\mathcal{N} \uparrow$.
- Define a volume-based diameter for each droplet, $d \equiv (6V/\pi)^{1/3}$.

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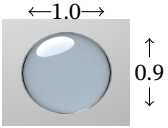
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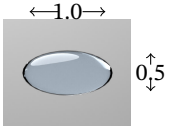
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- $d < d_H$: coalescence.
- $d > d_H$: breakup[b].

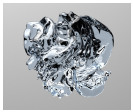
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What shape are the droplets?



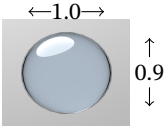




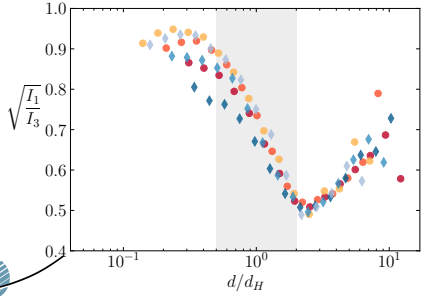
Aspect ratio $\sqrt{I_1/I_3}$	0.946	0.633	0.905
Sphericity d/d_A	0.999	0.965	0.701

- I_1 and I_3 are the smallest and largest moments of inertia.
- Area-based diameter for each droplet, $d \equiv \sqrt{A/\pi}$.

What shape are the droplets?

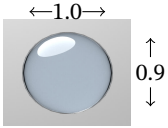


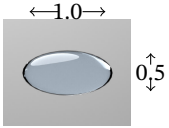
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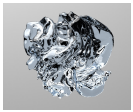


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- Area-based diameter for each droplet, $d \equiv \sqrt{A/\pi}$.
- $d << d_H$ aspect ratio $\rightarrow 1$.
- $d \approx d_H$ droplets are extended.
- $d >> d_H$ aspect ratio $\rightarrow 1$.

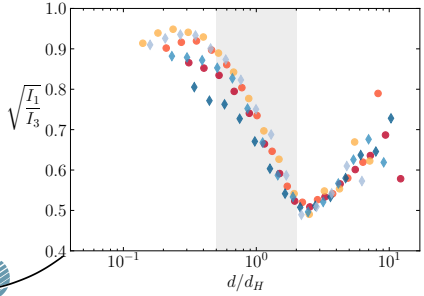
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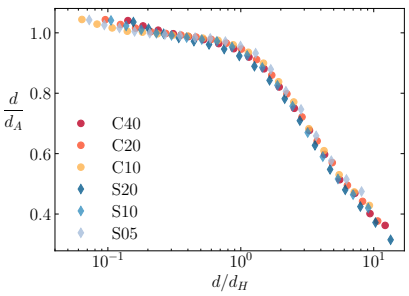






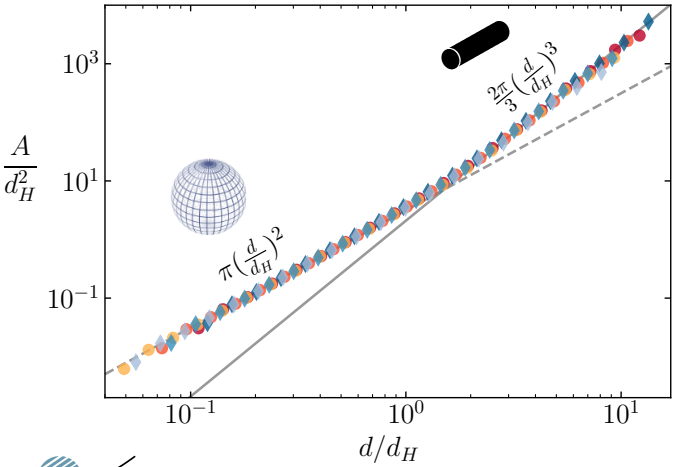
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- $d > d_H$ area $\uparrow$.

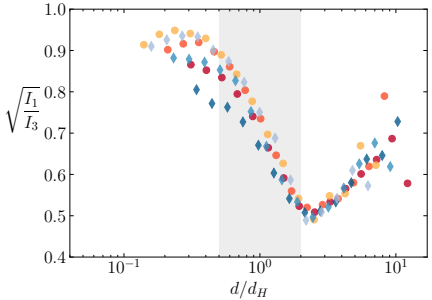
What is the area of the droplets?



- $d < d_H$: spheroids.
- $d > d_H$: filaments with diameter d_H.

Recap

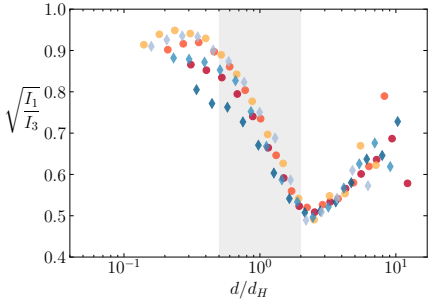
- The large droplets are filamentous.



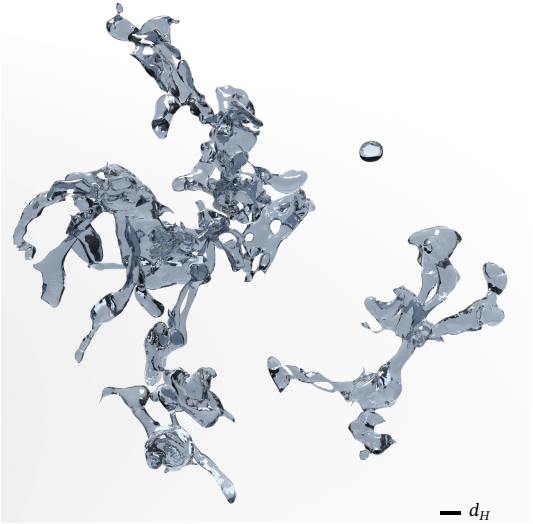
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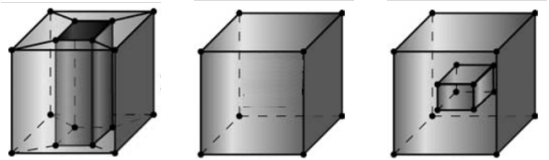
- Aspect ratio $\to 1$ implies the filaments must be coiled.



How to count the number of handles?

We compute the Euler characteristic[i] of the droplet interfaces.

$$\chi = n - e + f \tag{5}$$



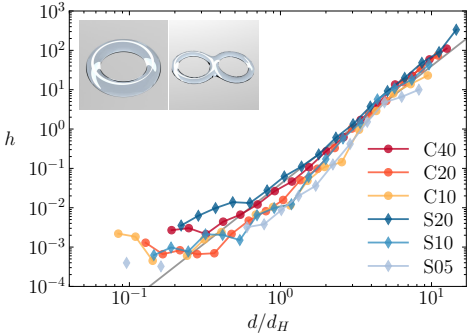
$\chi = 0$ $\chi = 2$ $\chi = 4$

Thus obtaining[j] the number of handles h and voids v.

$$\chi = 2 + 2v - 2h. \tag{6}$$

[i] Massey, W. S. (1997). Springer.
[j] Mendoza, R., Thornton, K., Savin, I., & Voorhees, P. (2006). *Acta Mater.*, (3).

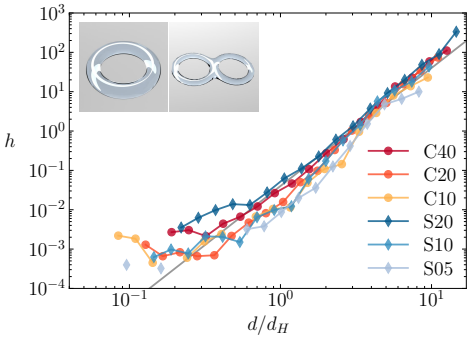
How many handles and voids are there?



$$h = 0.04(d/d_H)^3$$

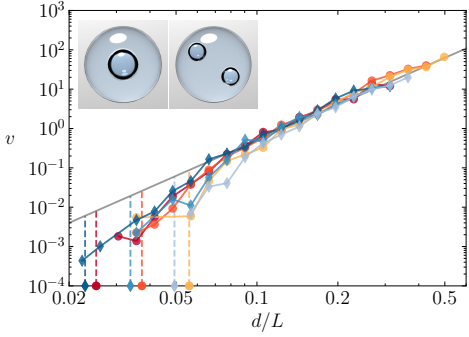
- Constant number of handles per unit length of filament.

How many handles and voids are there?



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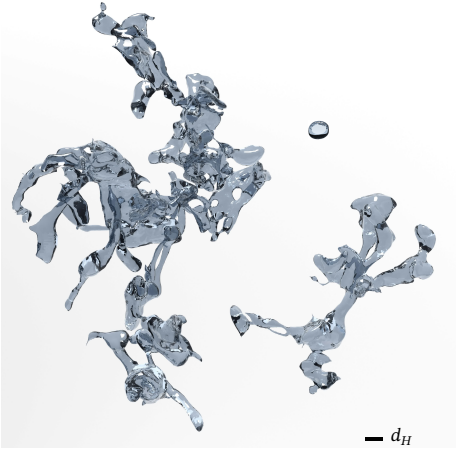


$v = 500(d/L)^3$

- Constant number of voids per unit volume of droplet.

Conclusions

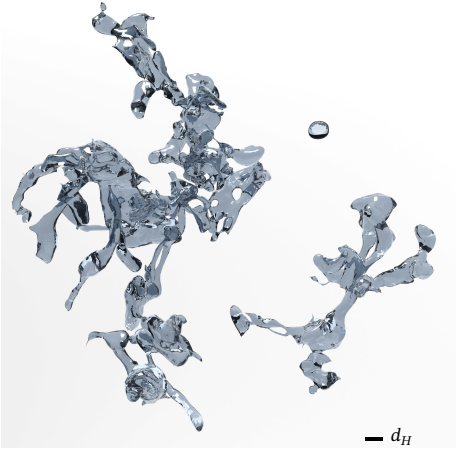
- Average surface tension can be used to define the Hinze scale d_H in surfactant-laden flows.



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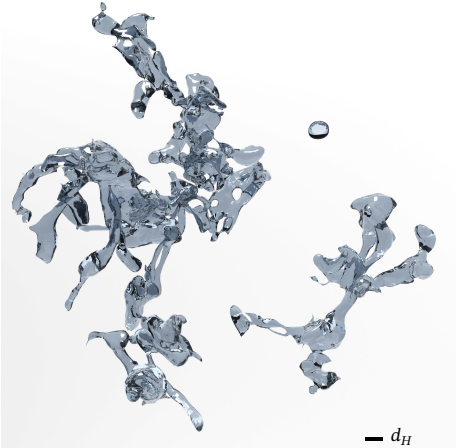
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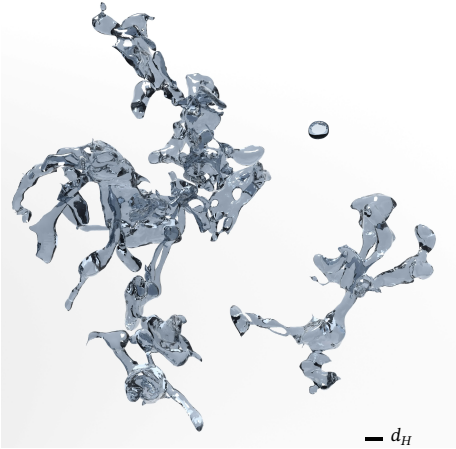


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Thank you!



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Bibliography I

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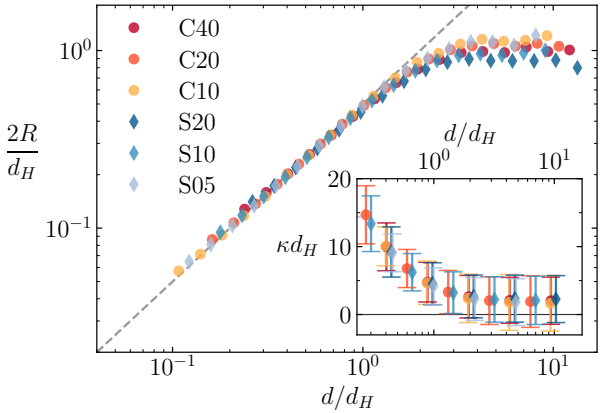
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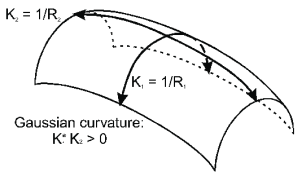
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Interface curvature

Curvature $\kappa = \kappa_1 + \kappa_2 = 1/R$.





- $d < d_H$: $R = d/4$.
- $d > d_H$: $R = d_H/2$, with regions of negative curvature.